

THE HONG KONG UNIVERSITY OF SCIENCE & TECHNOLOGY
ARIN5103: Foundations of Machine Learning
Fall 2025 Assignment 1

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Q1: Representation using perceptrons

First, we create a truth table for this expression.

a	b	c	$\neg a$	$\neg c$	$\neg a \vee b \vee \neg c$
0	0	0	1	1	1
0	0	1	1	0	1
0	1	0	1	1	1
0	1	1	1	0	1
1	0	0	0	1	1
1	0	1	0	0	0
1	1	0	0	1	1
1	1	1	0	0	1

A ONE perceptron computes a weighted sum of inputs plus a bias, then applies an activation function to produce an output (0 or 1). For inputs a, b, c , the weighted sum is:

$$z = w_a \cdot a + w_b \cdot b + w_c \cdot c + b_0$$

For $b = 1$, the results are all *True*; z should increase, so we set $w_b = 1$.

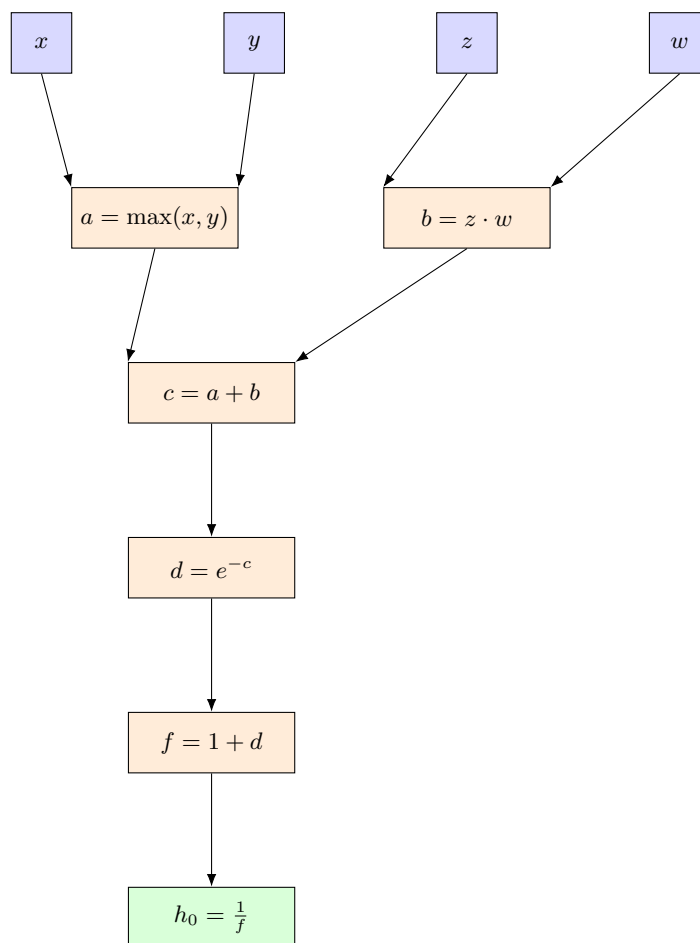
For $a, c = 1$; z should decrease, so we set $w_a = w_c = -1$.

And then if we use a **Heaviside step function**, b_0 should be 1.

The final answer is :

- **Inputs:** Boolean variables $a, b, c \in \{0, 1\}$
- **Weights:** $w_a = -1, w_b = 1, w_c = -1$
- **Bias term:** $b_0 = 1$
- **Weighted sum:** $z = w_a \cdot a + w_b \cdot b + w_c \cdot c + b_0$
- **Activation function:** Heaviside step function: $\text{output} = \begin{cases} 1 & \text{if } z \geq 0, \\ 0 & \text{if } z < 0. \end{cases}$

Q2: Computation graph



Calculate with $x = 1$, $y = -2$, $z = 0.5$, $w = 2$:

1. $a = \max(x, y)$: $a = \max(1, -2) = 1$
2. $b = z \cdot w$: $b = 0.5 \cdot 2 = 1$
3. $c = a + b$: $c = 1 + 1 = 2$
4. $d = e^{-c}$ (where $e \approx 2.71828$): $d = e^{-2} \approx 0.1353$
5. $f = 1 + d$: $f \approx 1 + 0.1353 = 1.1353$
6. final $h_0 = \frac{1}{f}$: $h_0 \approx \frac{1}{1.1353} \approx 0.8808$

The gradients of h_0 used in this:

1. $\frac{\partial h_0}{\partial f}$: From $h_0 = f^{-1}$, use power rule: $\frac{\partial h_0}{\partial f} = -f^{-2} = -\frac{1}{f^2}$
 When $f \approx 1.1353$: $\frac{\partial h_0}{\partial f} \approx -\frac{1}{(1.1353)^2} \approx -\frac{1}{1.2890} \approx -0.7769$
2. $\frac{\partial h_0}{\partial d}$: Since $f = 1 + d \implies \frac{\partial f}{\partial d} = 1$, by chain rule: $\frac{\partial h_0}{\partial d} = \frac{\partial h_0}{\partial f} \cdot \frac{\partial f}{\partial d} = -\frac{1}{f^2} \cdot 1$
 When $f \approx 1.1353$: $\frac{\partial h_0}{\partial d} \approx -0.7769 \cdot 1 \approx -0.7769$
3. $\frac{\partial h_0}{\partial c}$: Since $d = e^{-c} \implies \frac{\partial d}{\partial c} = -e^{-c} = -d$, by chain rule: $\frac{\partial h_0}{\partial c} = \frac{\partial h_0}{\partial d} \cdot \frac{\partial d}{\partial c} = -\frac{1}{f^2} \cdot (-d) = \frac{d}{f^2}$
 When $d \approx 0.1353, f \approx 1.1353$: $\frac{\partial h_0}{\partial c} \approx \frac{0.1353}{1.2890} \approx 0.1049$
4. $\frac{\partial h_0}{\partial a}$: Since $c = a + b \implies \frac{\partial c}{\partial a} = 1$, by chain rule: $\frac{\partial h_0}{\partial a} = \frac{\partial h_0}{\partial c} \cdot \frac{\partial c}{\partial a} = \frac{d}{f^2} \cdot 1$
 When $d \approx 0.1353, f \approx 1.1353$: $\frac{\partial h_0}{\partial a} \approx 0.1049$
5. $\frac{\partial h_0}{\partial b}$: Since $c = a + b \implies \frac{\partial c}{\partial b} = 1$, by chain rule: $\frac{\partial h_0}{\partial b} = \frac{\partial h_0}{\partial c} \cdot \frac{\partial c}{\partial b} = \frac{d}{f^2} \cdot 1$
 When $d \approx 0.1353, f \approx 1.1353$: $\frac{\partial h_0}{\partial b} \approx 0.1049$
6. $\frac{\partial h_0}{\partial x}$: Since $a = \max(x, y)$ and $x > y \implies \frac{\partial a}{\partial x} = 1$, by chain rule: $\frac{\partial h_0}{\partial x} = \frac{\partial h_0}{\partial a} \cdot \frac{\partial a}{\partial x}$
 When $x = 1$: $\frac{\partial h_0}{\partial x} \approx 0.1049$
7. $\frac{\partial h_0}{\partial y}$: Since $a = \max(x, y)$ and $y < x \implies \frac{\partial a}{\partial y} = 0$, by chain rule: $\frac{\partial h_0}{\partial y} = \frac{\partial h_0}{\partial a} \cdot \frac{\partial a}{\partial y}$
 When $y = -2$: $\frac{\partial h_0}{\partial y} = 0$
8. $\frac{\partial h_0}{\partial z}$: Since $b = z \cdot w \implies \frac{\partial b}{\partial z} = w = 2$, by chain rule: $\frac{\partial h_0}{\partial z} = \frac{\partial h_0}{\partial b} \cdot \frac{\partial b}{\partial z}$
 When $z = 0.5$: $\frac{\partial h_0}{\partial z} \approx 0.2098$
9. $\frac{\partial h_0}{\partial w}$: Since $b = z \cdot w \implies \frac{\partial b}{\partial w} = z = 0.5$, by chain rule: $\frac{\partial h_0}{\partial w} = \frac{\partial h_0}{\partial b} \cdot \frac{\partial b}{\partial w}$
 When $w = 2$: $\frac{\partial h_0}{\partial w} \approx 0.05245$